

Mandatory assignment 1

Introduction

You have been given the assignment to create a Linux service. You must choose one service to create, from several different options. You must also choose which programming language you will be using to create the service.

The service should have complete functionality, with options for starting and stopping using “systemd” or similar.

The service should be logging relevant details about its operations to the appropriate system logging directory.

Language

Bash is the subject of this course, and it is the preferred tool for the parts of the service that talk to the system: the unit file, process handling, file operations, and the command line utilities you have been taught.

You may also use Python where it genuinely earns its place. Parsing, keeping state between runs and working with structured data are all easier there, and you have far more hours of Python behind you than Bash.

Whatever you choose, state the split in your report and explain why you made it. If you write the whole thing in Python, explain why Bash was not the right tool (although this assignment is a great opportunity to learn Bash).

Levels

You must have completed the “Minimum” requirements and should aim to complete as many of the “Excellent” requirements as possible.

Your solution should also meet the requirements listed in “Requirements”. These must be documented in your report.

If you want to push even further, pitch your idea for a project! :)

Requirements

- **Robustness:**
Your service must be able to recover from a crash, start up on its own when the system is started.
- **Logging:**
Your service should output relevant logging details with timestamps, in case of a crash, restart, startup, or warnings during runtime.
- **Unit file:**
The service should have a unit file with configuration and relevant variables depending on the service of your choice.
- **Programming language:**
You must choose Bash or Python for your service.

Deliverables

- A .zip file containing:
 - o The source code for your service
 - o The unit file for your service
 - o An example logfile from the output of your service
- A short report and summary about what your service does and how it works, preferably with some screenshots. **Maximum 5 pages.**

Option 1: Log maintenance service

Monitor a directory of log files, compress them when they grow too large, and remove them when they get too old.

Minimum:

- Watch one directory. Compress any file above a configurable size.
- Delete any archive older than a configurable age.
- Log every action taken, with the filename and the reason for it.

Excellent:

A good solution handles these awkward cases:

- the directory does not exist
- a file is being written to while you work on it
- filenames contain spaces

Your solution should also:

- never compress the same file twice
- never delete something you created seconds earlier. Log a summary per run:

Instead of logging every action, log a summary after each “run”.

The summary should include:

- files seen
- files compressed
- files deleted
- bytes reclaimed (total size saved by compressing)

Option 2: Availability monitor

Check that a list of hosts, ports or URLs is reachable, and log when their state changes.

Minimum:

- Read a list of targets from the unit file or a config file
- Check each one on an interval
- Log each result with a timestamp and whether the target was reachable

Excellent:

A good solution only logs the *transitions* instead of each result:

- one line when a target goes down
- one line when it comes back (and nothing in between)

When logging a transition, a good solution should also:

- include the duration of the outage in the recovery line
- decide what to do about a target that is already unreachable when the service starts when you have no previous state to compare against, and log that on start up